HDOC Day-48

Question

Write a C program to read the size and elements of a variable-length integer array. Pass the array by address to pivot_index() using pointers. Display the leftmost index whose left sum equals its right sum, or -1 if none exists.

C Program

```
#include <stdio.h>

void pivot_index(int *arr, int n) {
    int total = 0, left = 0;
    for (int i = 0; i < n; i++)
        total += *(arr + i);

    for (int i = 0; i < n; i++) {
        int right = total - left - *(arr + i);
        if (left == right) {
            printf("%d\n", i);
            return;
        }
        left += *(arr + i);
    }
    printf("-1\n");
}

int main() {
    int n;
    scanf("%d", &n);
    if (n <= 0) return 0;
    int arr[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    pivot_index(arr, n);
    return 0;
}
```

Sample Input and Output

```
Input:
6
1 7 3 6 5 6

Output:
3
```

Explanation

The total sum is calculated first. For each index, right sum = total - left sum - current element. The first matching index is printed.